SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – INDIVIDUAL TALK

Course Code:	ITA0301	**Course Title:**	Mobile computing
CO Assessed:	CO1	**Assessment:**	Assessment Tool 3 – INDIVIDUAL TALK
Weightage:	30 % of CIA	**Total Marks:**	20 marks
CO1 Statement:	*Analyse the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.*		
Student Name:	Gangisetty Anjani Kumar	**Reg. No.:**	192373022
Section / Batch:	CSE-DS (4th year)	**Date:**	05-10-2026

Code of Conduct

I __Gangisetty Anjani Kumar (192373022)__ (Name / Reg. No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

*Signature of the Student: _G. Anjani Kumar_*

Instructions

- Topic given before the session and students speak on stage.
- Duration: 10 minutes (In class)

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Individual Talk:

Network Switching Subsystem (NSS) Operations

Introduction

The Network Switching Subsystem (NSS) is an important part of a cellular mobile communication network, especially in GSM-based networks.

The main purpose of the NSS is to manage the switching, routing, mobility, subscriber information, authentication, and call-related operations of mobile users.

The main components of NSS are MSC, GMSC, HLR, VLR, AuC and EIR.

MSC, or Mobile Switching Center, is the main switching component. It establishes and terminates calls, performs call routing and supports mobility-related operations.

GMSC, or Gateway Mobile Switching Center, acts as a gateway between the mobile network and external networks such as the public telephone network.

HLR, or Home Location Register, is the main subscriber database. It stores information about subscribers and their subscribed services.

VLR, or Visitor Location Register, temporarily stores information about subscribers who are currently present in a particular service area.

AuC, or Authentication Center, supports subscriber authentication and helps prevent unauthorized access to the network.

EIR, or Equipment Identity Register, stores information about mobile equipment identities such as the IMEI.

When a mobile user makes a call, the request goes through the **BSS to the MSC**. The NSS uses subscriber and location information to identify the destination and route the call correctly.

NSS also supports **mobility management**. When a user moves from one cellular area to another, the network updates the user's location and helps maintain communication.

It also supports **roaming**, allowing users to receive mobile services outside their home network area when roaming is supported.

1. **Call Switching:** Call Switching means connecting a call from one user to another user through the mobile network.
2. **Call Routing:** Call Routing means deciding the correct path through which a call should travel to reach the destination user.
3. **Subscriber Management:** Subscriber means a person who uses a mobile network service.
4. **Authentication:** Authentication means checking whether a mobile user is an authorized user or not.
5. **Mobility Management:** Mobility Management means managing a mobile user when they move from one location/cell to another.
6. **Roaming:** Roaming means using mobile network services when you are outside your home network/service area, subject to service availability and operator agreements.

Conclusion

To conclude, the Network Switching Subsystem (NSS) is an important part of a GSM cellular network. It manages call switching, call routing, subscriber information, authentication, mobility, and roaming. The main NSS components such as MSC, GMSC, HLR, VLR, AuC, and EIR work together to provide reliable and continuous mobile communication. Therefore, NSS plays a key role in managing the core operations of a mobile network.

Geotag Photo



Thank you